

ZenVoice FluidAudio Removal Report

Date: 2026-08-05

Status: Phase 1 completed

1. Executive Summary

ZenVoice previously depended on the closed-source `FluidAudio` Swift package to run the NVIDIA Parakeet Unified English CoreML model. Because FluidAudio is a proprietary competitor dependency, and because ZenVoice is moving to a fully open-source Apache-2.0 project, every FluidAudio reference has been removed.

The **Parakeet model weights themselves** are not legally bound to FluidAudio.

They are published by NVIDIA on Hugging Face under the NVIDIA Open Model License.

What **is** bound to FluidAudio is the **inference engine code** that knows how to load and execute the converted CoreML `.mlmodelc` bundle.

This report documents:

- exactly which FluidAudio components ZenVoice was using,
- which files were changed or deleted,
- which models are now retired vs. reusable,
- and the legal boundary between FluidAudio code and NVIDIA/Parakeet weights.

2. What was FluidAudio providing?

FluidAudio was a Swift package hosted at

``https://github.com/FluidInference/FluidAudio.git``, pinned in ZenVoice's

``Package.swift`` at revision ``88d6d8166880dee1ac7c32c80f8e10cd782f8ca8``.

ZenVoice used exactly one product from that package:

| ZenVoice file | Imported symbol | Purpose

| ``Sources/ZenVoiceRuntime/ParakeetModelSupport.swift`` | ``FluidAudio.UnifiedAsrManager`` | Load the Parakeet CoreML bundle and run inference on 16 kHz mono audio samples.

``UnifiedAsrManager`` is a closed-source class. It wrapped the following work:

- loading the Parakeet Unified EN CoreML `.mlmodelc` directory,
- running the encoder (INT8 FastConformer),

- running the decoder + joint-decision transducer loop,
- returning text and per-token timings.

ZenVoice did **not** use any other FluidAudio product (no text-normalization runtime, no speaker diarization, no clustering wrapper) in its current code.

Those extra components were only noticed in `THIRD\_PARTY\_NOTICES.md` because they were compiled into the `FluidAudio` binary and therefore redistributed with ZenVoice.

3. What Parakeet / NVIDIA assets were involved?

| Asset | Source | License | Bound to FluidAudio?

| `nvidia/parakeet-unified-en-0.6b` base weights | NVIDIA on Hugging Face | NVIDIA Open Model License | **No**

| `FluidInference/parakeet-unified-en-0.6b-coreml` CoreML conversion | FluidInference on Hugging Face | Declared CC-BY-4.0, but artifact metadata says NVIDIA Open Model License | **No** (it is a model file; the license applies to the weights, not to FluidAudio code)

| `FluidAudio.UnifiedAsrManager` runtime | `FluidInference/FluidAudio` | Closed-source / proprietary | **Yes** this is the only thing ZenVoice could not legally redistribute

The Parakeet model was offered in ZenVoice under the catalogue ID

`parakeet-unified-en-int8`. Its runtime metadata was `.parakeetCoreML`. The model was a multi-file directory (`parakeet-unified-en-0.6b`) containing:

- `config.json`
- `metadata.json`
- `vocab.json`
- `parakeet\_unified\_decoder.mlmodelc`
- `parakeet\_unified\_encoder\_int8.mlmodelc`
- `parakeet\_unified\_joint\_decision\_single\_step.mlmodelc`

ZenVoice downloaded these files file-by-file from Hugging Face, verified them against a pinned manifest, and then handed the local directory to `UnifiedAsrManager.loadModels(from:)`.

4. Which files were changed or deleted, and why?

4.1 Deleted files

| File | Why deleted

| `Sources/ZenVoiceRuntime/ParakeetModelSupport.swift` | Entire file was glue code to `FluidAudio.UnifiedAsrManager`. It contained no model weights and no independent inference logic. It could not exist without the FluidAudio import.

| `Package.resolved` | SwiftPM regenerates this file. Deleting it forced a clean resolution that no longer includes FluidAudio.

4.2 Modified files code

| File | What changed

| `Package.swift` | Removed the `FluidAudio` dependency and the `.product(name: "FluidAudio", package: "FluidAudio")` product from `ZenVoiceRuntime`.

| `Sources/ZenVoiceRuntime/WhisperTranscriber.swift` | Removed the `parakeetEngine` property, the `transcribeWithParakeet` method, and all Parakeet warm-up branching. All transcription now goes through `whisper.cpp`.

| `Sources/ZenVoiceCore/VerifiedModelCatalog.swift` | Removed `SpeechModelRuntime.parakeetCoreML`, `VerifiedModelFile`, bundle URL logic, manifest digest, and bundle verification. Moved `parakeet-unified-en-int8` from active `models` to `retiredModels`.

| `Sources/ZenVoiceCore/ModelRecommendations.swift` | English recommendation on Apple Silicon changed from `parakeet-unified-en-int8` to `whisper-large-v3-turbo`.

| `Sources/ZenVoice/ModelManagerViewModel.swift` | Removed the `downloadParakeet` and `downloadBundleFile` methods. All downloads are now single-file GGML downloads.

| `Sources/ZenVoiceCoreChecks/main.swift` | Updated model counts, recommendation assertions, and removed bundle-verification tests.

4.3 Modified files license and documentation

| File | What changed

| `LICENSE` | Replaced proprietary/source-visible license text with Apache License, Version 2.0.

| All `Sources/\*\*/\*.swift` | Added Apache-2.0 license header (mechanical change).

| `THIRD\_PARTY\_NOTICES.md` | Removed FluidAudio, fastcluster, VBx, and NemoTextProcessing notices. Kept `whisper.cpp`, OpenAI Whisper weights, and Hinglish Apex weights. Added a "Retired NVIDIA Parakeet" notice explaining the model is no longer used.

| `README.md`, `docs/\*.md`, `CHANGELOG.md` | Updated every reference that described Parakeet/FluidAudio as active. Replaced with `whisper.cpp`-only descriptions and Apache-2.0 project status.

5. Why the Parakeet model was retired, not deleted

The `parakeet-unified-en-int8` entry was moved to `retiredModels` instead of

being removed entirely. This is a **user-safety** decision, not a legal one.

- ZenVoice stores the user's selected model by catalogue ID in `UserDefaults`.

- If a user had already downloaded and selected Parakeet, deleting the catalogue entry would make that installed model become "no model installed" on next launch.

- A retired entry stays resolvable, so the app can still display it, verify it, and offer to remove it but it is no longer offered for new downloads.

6. Can ZenVoice use Parakeet models again in the future?

Yes. The model weights are not bound to FluidAudio. To re-offer Parakeet,

ZenVoice must provide its own open-source inference engine. Realistic paths:

| Path | Description | Complexity

| A. Native CoreML loader | Write Swift code that loads the existing `.mlmodelc` encoder, decoder, and joint-decision models directly with `CoreML.MLModel` and implements the transducer beam loop. | High the bundle has three interdependent models and a custom tokenizer.

| B. NeMo / ONNX bridge | Convert or run the Parakeet checkpoint using NVIDIA NeMo or an ONNX runtime, invoked from Swift. | Medium-High adds a runtime dependency and conversion step.

| C. Whisper-only for now | Keep Parakeet retired and focus on better Whisper models (turbo, distillation, fine-tunes) that run on `whisper.cpp`. | Low current path.

None of these paths require FluidAudio's permission, because none of them use FluidAudio's source code.

7. What models are legally reusable today?

| Model | Publisher | License | Reusable?

| Whisper Tiny/Base/Small/Medium/Large/Turbo (GGML) | OpenAI weights, ggml-org conversion | MIT | **Yes** already in use.

| Whisper-Hindi2Hinglish-Apex (GGML) | Oriserve weights, ZenVoice conversion | Apache-2.0 | **Yes** already in use.

| Parakeet TDT / Unified / Realtime weights | NVIDIA | CC-BY-4.0 / NVIDIA Open Model License | **Yes** weights can be downloaded and used with any engine.

| Parakeet `.mlmodelc` conversion by FluidInference | FluidInference | CC-BY-4.0 declared | **Yes** as a model file, but ZenVoice has no open engine for it.

| FluidAudio runtime code | FluidInference | Proprietary/closed-source | **No** removed.

8. Legal conclusion

ZenVoice's removal of FluidAudio was necessary because:

1. The `FluidAudio` Swift package is closed-source/proprietary.
2. An Apache-2.0 project cannot ship a binary dependency whose license does not permit redistribution under Apache-2.0 terms.
3. The only part of the Parakeet pipeline that had this problem was the **FluidAudio inference engine**, not the **NVIDIA model weights**.
4. The Parakeet model was therefore retired from active use until ZenVoice has its own open-source engine for it.

The NVIDIA Parakeet models remain available on Hugging Face and can be reused in ZenVoice the moment the project builds or integrates a compatible open-source runtime.

9. Verification

After removal, the project builds and all checks pass:

<code>swift build</code>	Build complete
<code>swift run ZenVoiceCoreChecks</code>	all checks passed
<code>swift run ZenVoiceRuntimeChecks</code>	Whisper path passed
<code>swift run ZenVoiceStorageChecks</code>	17 checks passed

The only local speech runtime now embedded in ZenVoice is the

`whisper.cpp` v1.9.1 XCFramework, licensed under MIT.